On $(3, 10, 40)$-graphs: structure, symmetry, and certified computations

A progress report

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Progress report. This document records the progress of an ongoing project. It is not a formal paper and gives no definite answer: $R(3, 10)$ remains open, and the results are partial. Each result carries its status (Sec. 2.2); work in progress is listed in Sec. 10.

1 Introduction

The Ramsey number $R(3, 10)$ satisfies $40 \leq R(3, 10) \leq 41$. The lower bound comes from Exoo's $(3, 10, 39)$-graph [Ex89]. Goedgebeur and Radziszowski proved $R(3, 10) \leq 42$ by neighbourhood gluing [GR], and Angeltveit improved this to $R(3, 10) \leq 41$ [Ang] (arXiv:2401.00392). So $R(3, 10) = 40$ if and only if no $(3, 10, 40)$-graph exists: a triangle-free graph on 40 vertices with independence number at most 9. The question is open. This progress report gives structural results on a hypothetical $(3, 10, 40)$-graph G and give an exact status for each result.

Our results (status labels as in Sec. 2.2; published inputs listed in Sec. 2.3):

1. Structure (Sec. 3): degrees 6..9 after the WLOG reduction to maximal triangle-free graphs, $\Delta = 9$, codegree at most 7, Theorems A and A$''$ (averaging), 545 possible MTF degree distributions, and lemmas for the 9-regular case.
2. Formal verification (Sec. 4): the hand lemmas and the case tables of Sec. 3 are checked in Lean 4, with the published facts as explicit hypotheses.
3. Symmetry (Sec. 5): G is not a Cayley graph (all 14 groups of order 40, certified), and has no automorphism of several prime orders and fixed-point counts (certified or proved; order 5 with 5 fixed points is enumeration only). Order 2 with $a = 7$ is closed except $f = 34$: by proof for the empty f values, and certified for the others (2557 SAT runs, all DRAT VERIFIED, see Sec. 5.5).
4. Gluing (Sec. 6, enumeration, re-checked on samples with independent code, Review 31): a degree-8 vertex v has $e(G_v) \geq 97$, and a degree-9 vertex v has $e(G_v) \geq 88$. This closes 20 of the 545 distributions; 525 stay open, including the 9-regular case.
5. Certified SAT (Sec. 7): SMS with a lazy independent-set propagator and proof logging; certified small edge bounds; for the 9-regular sub-case $R(7)$, 365 of 464 cubes are certified UNSAT with a certified cover.
6. Negative results (Sec. 8): we measured every route we found for the remaining cases (cube-and-conquer, gluing, edge kernels, LP relaxations, literature methods). By our estimates, none of these routes closes the remaining cases within a 180 CPU-year budget. For case R, the main

routes are estimated at 10^2 to 10^7 times the budget (statistical estimates from random probes). We give the measurements so that others need not repeat them.

Sec. 9 lists the released data and code.

2 Preliminaries and published facts used

2.1 Notation

All graphs are simple. A $(3,k;n)$-*graph* is a triangle-free graph on n vertices with independence number $\alpha \le k-1$; $R(3,k;n,e)$ is the set of $(3,k;n)$-graphs with e edges (up to isomorphism), and $e(3,k,n)$ is the least e for which it is non-empty (infinite if empty). A $(3,10,40)$-*graph* G is a $(3,10;40)$-graph; we assume one exists and derive structure. $d(v)$ is the degree, n_d the number of vertices of degree d, $e = e(G)$, $N(v)$ the neighbourhood, $N[v] = N(v) + v$, and

$$G_v = G - N[v] \quad (39 - d(v) \text{ vertices; also written } H_v \text{ in the notes}), \qquad Z(v) = \sum_{u \in N(v)} d(u).$$

The *deficiency* is $N = \sum_v (9 - d(v)) = 360 - 2e$. For non-adjacent u, w the *codegree* is $c(u,w) = |N(u) \cap N(w)|$. G is *maximal triangle-free* (MTF) if adding any edge creates a triangle, i.e. every non-adjacent pair has a common neighbour.

2.2 Status labels

Every statement in Sections 3–4 carries one of these labels.

- [pub] published result, used as a hypothesis (ID in Sec. 2.3; not re-verified by us).
- [hand] hand proof in this report, independently re-checked ("CONFIRMED", with the review number in the review log `notes/critic.md`; Sec. 10).
- [Lean] proved in Lean 4 / Mathlib, with the published facts it uses as explicit hypotheses (Sec. 4).
- [cert] a computation whose result is checked by a verified checker (exact Farkas certificates in rational arithmetic, or SAT proofs checked by cake_lpr / VeriPB, Sec. 7).
- [enum] an exhaustive enumeration without a proof certificate (gluing runs, Sec. 6).
- [cond: X] proved under the additional hypothesis X.

2.3 Published facts

IDs are those of our dependency list (`notes/DEPENDENCIES.md`). We use only results published in journals or on arXiv. [GR] = Goedgebeur, Radziszowski; [Ang] = Angeltveit; [McZ] = McKay, Zhang; [GrR] = Grinstead, Roberts; [DuLBG] = Li, Duggan, Bright, Ganesh.

ID	statement	source	used in
D-R1	$R(3,k) = 6, 9, 14, 18, 23, 28, 36$ for $k = 3..9$	[GR] Table 1	Lemma H (3.8)
D-R2	$R(3,8) = 28$	[McZ]; DRAT-certified in [DuLBG]	L6, Thm A″, Cover Lemma ("F2")
D-R3	$R(3,9) = 36$	[GrR]; certified in [DuLBG]	L1
D-R4	$40 \le R(3,10) \le 41$	Exoo 1989 [Ex89]; [Ang] Thm 1	Sec. 1
D-E8	$e(3,8,n) = 25, 30, 35, 42, 49, 56, 65, 73, 85$ for $n = 19..27$	[GR] Tables 3, 13	Lemmas P, R7, F
D-E9	$e(3,9,n) = 86, 95, 104, 118, 129, 140$ for $n = 30..35$	[GR] Tables 3, 4	L3, Thm A, case tables
D-E10b	$e(3,10,40) \ge 161$	[GR] Table 5	$\Delta = 9$, case tables ("GR161")
D-G1	$R(3,9;35) = \{C_{35}(1,7,11,16)\}$, 8-regular, 140 edges	[GR] Thm 3	L7 (degree-4 vertices)
D-C8d	$\lvert R(3,8;26,e)\rvert = 62, 1625, 23409, 216151, 1526296$ for $e = 73..77$ (1,767,543 with $e \le 77$)	[GR] Table 13; [Ang] Sec. 3.2	Lemma F, $R(6)$ branch
D-C8e	$\lvert R(3,8;27,85)\rvert = 4$	[GR] Table 13	Lemma R7
D-C9a–e	$\lvert R(3,9;30,86)\rvert = 1800$, $\lvert R(3,9;30,87)\rvert =$ 147,335; $\lvert R(3,9;31,95/96)\rvert = 560/35{,}154$; $\lvert R(3,9;32,\le 112)\rvert =$ 1,554,928,360; $\lvert R(3,9;33,\le 121)\rvert =$ 14,395	[GR] Table 14; [Ang] Sec. 3.3	gluing scenarios (3.5, Sec. 6)

2.4 Two base facts and their status

The coarse case split (Cover Lemma, Sec. 3.6) uses only:

- **F1**: $e(3,8,m) \ge 6m - 91$ for $23 \le m \le 27$. [pub] D-E8 (true values $49, 56, 65, 73, 85 \ge 47, 53, 59, 65, 71$). In addition [cert] for $m = 23, 24, 27$ (SAT, our runs, Sec. 7); $m = 25, 26$ rest on D-E8 only.
- **F2**: $R(3,8) \le 28$. [pub] D-R2, with the DRAT certificate of [DuLBG].

The refined case tables (Sec. 3.5) also use D-E9 and D-E10b.

3 Structure of a $(3, 10, 40)$-graph

Throughout, G is a $(3, 10, 40)$-graph. Lean names refer to Sec. 4 (namespace `R310`).

3.1 Degrees and edges

L1. $4 \le d(v) \le 9$.

$N(v)$ is independent, so $d(v) \le 9$; G_v has $\alpha \le 8$, so $39 - d(v) \le R(3,9) - 1 = 35$. [hand, Lean `degree_mem_Icc`; uses D-R3]

L2–L4. *$e(G_v) = e - Z(v)$; $e(G_v) \ge e(3, 9, 39 - d(v))$; $\sum_v Z(v) = \sum_v d(v)^2$, hence $\sum_v e(G_v) = \sum_v d(v)(20 - d(v))$.*

[hand, Lean `card_edgeFinset_delNbhd_add_sum_degree`, `e39_le_card_edgeFinset_delNbhd`, `sum_card_edgeFinset_delNbhd_add_sq`; uses D-E9]

L5. $159 \le e \le 180$ *[hand, Lean `le_card_edgeFinset`, `card_edgeFinset_le_180`; uses D-E9];* $e \ge 161$ *[pub D-E10b]. Hence* $\Delta(G) = 9$*:* $2e \ge 322 > 320 = 40 \cdot 8$ *[Lean `exists_degree_nine`, hyp. D-E10b].*

L6 (pair lemma). *For non-adjacent* $u \ne w$*:* $c(u,w) \le d(u) + d(w) - 11$*; in particular* $c(u,w) \le 7$*, and* $d(u) + d(w) \le 10$ *forces* $u \sim w$*.*

Proof. $X = V - N[u] - N[w]$ has $\alpha \le 7$, so $|X| = 38 - d(u) - d(w) + c \le 27$ (F2). □

Consequences: $n_4 + n_5 \le 2$, and further restrictions on degree-4 vertices. [hand, Lean `pair_lemma`, `adj_of_degree_add_le`, `card_degree_le_five`; uses F2]

L7 (degree 4). *If* $d(u) = 4$ *then* G_u *is the unique* $(3, 9; 35)$*-graph, every vertex outside* $N[u]$ *has degree 8 or 9,* $e = 140 + Z(u)$*, and any two neighbours of* u *have degree sum* ≥ 12*.*

[hand, Lean `deg4_*`; uses D-G1] A separate SAT run shows $\delta(G) \ge 5$ [cert: drat-trim + cake_lpr, `notes/critic.md` Reviews 4–5; uses D-G1]; it is not needed below, because MTF graphs have $\delta \ge 6$ (M1).

3.2 Maximal triangle-free graphs

M0. *If a* $(3, 10, 40)$*-graph exists, an MTF* $(3, 10, 40)$*-graph exists (add edges while no triangle appears;* α *does not increase).*

[hand, Lean `exists_isMTF_le`]

M1. *In an MTF* $(3, 10, 40)$*-graph,* $\delta \ge 6$*.*

Proof. Let $d = d(v)$, $A = N(v)$, $H = G_v$. Every vertex of H has a neighbour in A; those with exactly one neighbour x in A, together with $A - x$, are independent, so there are at most $d(10 - d)$ of them. Hence $8d \ge e(A, H) \ge 2(39 - d) - d(10 - d)$, i.e. $d(20 - d) \ge 78$, false for $d \le 5$. □

[hand, Lean `IsMTF.six_le_degree`; uses no published fact]

M2. *In an MTF graph,* $Z(v) \ge d^2 - 11d + 78$ *(*$= 48, 50, 54, 60$ *for* $d = 6..9$*).*

[hand, Lean `IsMTF.sum_degree_neighbor_ge`]

M3. *A 9-regular* $(3, 10, 40)$*-graph is MTF (an added edge would give degree 10).*

[hand, Lean `isMTF_of_regular_nine`]

Scope. M0 is valid only for plain existence. A search that also assumes an automorphism (Sec. 5) or any other property that adding edges can destroy must use the lists for all G, not the MTF lists.

3.3 Averaging: Theorem A

Theorem A. *G has a vertex v with ($d(v) = 9$ and $e(G_v) \le 99$) or ($d(v) = 8$ and $e(G_v) \le 96$). If G is not 9-regular, the same holds with 98 in place of 99.*

Proof. Put $a_v = e(G_v) - d(v)(20 - d(v))$; $\sum_v a_v = 0$ (L4). For $d \le 7$, $e(3, 9, 39 - d) > d(20 - d)$, so $a_v \ge 13$. For the second claim: if no vertex is as stated, $a_v \ge 0$ at 9-vertices and $a_v \ge 1$ at 8-vertices, so $\sum_v a_v > 0$ unless G is 9-regular. For the first claim: if no vertex is as stated, $a_v \ge 1$ at every 9-vertex ($e(G_v) \ge 100$) and at every 8-vertex, so $\sum_v a_v > 0$ in all cases. □

[hand, CONFIRMED; Lean `theoremA`, `theoremA_of_not_regular`; uses D-E9]

Theorem A″. *For MTF G the table input of Theorem A reduces to $e(3,9,32) > 91$ and $e(3,9,33) > 84$, and these follow from F1 and F2 (Jensen's inequality applied to $\sum_w q_M(d_H(w)) \geq 0$ for H in $R(3,9;M)$, $M = 32, 33$, gives $e(H) \geq 95$ and $e(H) \geq 103$).*

[hand, CONFIRMED Review 16; Lean `thmA2_32`, `thmA2_33`, `theoremA_mtf`] Theorems B and C (Angeltveit-type averaging and a closed form for degrees $\{8, 9\}$) are also proved [hand, Lean `theoremB`, `theoremC`], but they give no smaller classes than the LP of Sec. 3.4.

3.4 Degree distributions

`cases/degseq.py` enumerates the 1,221,759 tuples $(n_4, \ldots, n_9)$ with sum 40 and removes those that violate parity, L5, M1 (MTF only), the sum inequality $\sum_d n_d(d(20-d) - e(3,9,39-d)) \geq 0$ (from L4), L6, L7, M2, or the *class LP*. The class LP has variables m_{ij} = number of edges between degree classes C_i and C_j, with rows for degree sums, $\sum_{v \in C_i} Z(v) \leq n_i(e - L_i)$, weighted Mantel (for every class weight vector in $\{0..3\}^{\text{classes}}$), simple caps, L6, and M2 (MTF). A distribution is removed by the LP only with an exact Farkas certificate.

mode	survivors	removed by the LP
all G, $e \geq 161$ (D-E10b)	1897	47
all G, $e \geq 159$ (L5 only)	1938	51
MTF G, $e \geq 161$	**545**	0
MTF G, $e \geq 159$	573	0

MTF ranges: $n_4 = n_5 = 0$, $n_6 \leq 9$, $n_7 \leq 19$, $n_9 \geq 2$, $161 \leq e \leq 180$. All 19,890 certificates were re-checked by an independent script, and the four lists are checked in Lean from pinned outputs (`caseTable_<tag>`, Sec. 4.3). [cert + Lean; CONFIRMED Reviews 6, 9]

Per-distribution bound t_9. For a set U of excluded classes ("no vertex of degree d has $e(G_v) \leq U_d$"), $t_9(n)$ is the least T such that "every 9-vertex has $e(G_v) \geq T + 1$" is LP-infeasible (with certificate) for the distribution n. So every graph with distribution n has a 9-vertex v with $e(G_v) \leq t_9(n)$. Over the 545 MTF distributions with $U_8 = 96$, t_9 takes the values 85..99 with multiplicities 85: 6, 86: 8, 87: 6, 88: 15, 89: 24, 90: 43, 91: 59, 92: 80, 93: 91, 94: 85, 95: 74, 96: 37, 97: 14, 98: 2, 99: 1. [Lean `t9_gr_mtf_U8_main` and the other scenarios, hypothesis `Glue U G`]

3.5 What the done gluing runs close, and what stays open

Sec. 6 reports two exhaustive gluing runs [enum]: no degree-8 vertex has $e(G_v) \leq 96$ (all 35,714 graphs of $R(3,9;31,\leq 96)$), and no degree-9 vertex has $e(G_v) \leq 87$ (all 149,135 graphs of $R(3,9;30,\leq 87)$). With these, the 20 MTF distributions with $t_9 \leq 87$ are impossible (all 20 with $e = 161$ among them). **525 MTF distributions stay open: the 9-regular one (case R) and 524 with $162 \leq e \leq 179$.** [cond: the two gluing runs, enum; runs CONFIRMED Review 31] A $t_9 \leq 88$ gluing with no output closes 15 more (510 open; Sec. 10). These two runs also make $e \geq 161$ unnecessary for the MTF table: the 28 extra distributions with $e \in \{159, 160\}$ are closed by them (Review 21).

Per edge count (MTF, open after the two runs; $t_9^{\max}$ = largest t_9 among them):

e	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
open	19	18	17	16	80	70	61	52	44	37	30	24	19	14	10	7	4	2	1
$t_9^{\max}$	90	92	93	94	94	95	96	97	97	96	96	96	96	96	97	97	97	98	99

A conditional reduction ("sel"). If, in addition, no vertex had $e(G_v) \le U_d$ for $(U_6, U_7, U_8, T^*) = (121, 112, 99, 90)$, only 208 MTF distributions would remain (case R and 207 with $e \ge 167$); a profile ILP removes 7 more, with exact Farkas certificates (7/7 checked), leaving 200 + R [cond: sel; cert for the 7; CONFIRMED Review 27]. The hypothesis sel is a gluing statement over input classes of $\sim 5 \cdot 10^9$ graphs (MCMC estimates, Sec. 6), about 100 times beyond the speed of our gluer. The 200/208 lists are therefore conditional on sel.

3.6 Cover Lemma (the coarse split used for certified SAT)

Let Φ = "triangle-free and $\alpha \le 9$" on $V = \{0..39\}$, and MTFc = "every non-adjacent pair has a common neighbour".

cube	constraints added to Φ
R	every degree is 9; $N(0) = \{1..9\}$
S8	MTFc; $d(0) = 8$; $N(0) = \{1..8\}$; $e(G_0) \le 96$
N	MTFc; $d(0) = 9$; $N(0) = \{1..9\}$; $e(G_0) \le 98$

Cover Lemma. *Assume F1, F2. If a $(3, 10, 40)$-graph exists, some graph on V satisfies Φ, one of R, S8, N, and $c(u, w) + 11 \le d(u) + d(w)$ for all non-adjacent u, w.*

Proof. M0, M1, then Theorem A (via A″) and a relabelling; L6 for the codegree rows. □

[hand, CONFIRMED; Lean `cover_lemma`, clause form `Enc.cover_assign`, file form `Files.cover_files`; hypotheses only F1, F2] Symmetry breaking is valid in each cube if its permutations fix 0 and the blocks $N(0)$ and $V - N[0]$ as sets (Lean `Files.cover_files_sb`, Sec. 4.3). Cube S8 is closed by the S8 gluing run [enum]; a certified SAT proof is open.

3.7 The 9-regular case R

L9. *If G is 9-regular: $e = 180$, every G_v is in $R(3, 9; 30, 99)$, and for non-adjacent u, w: $1 \le c(u, w) \le 7$. Every vertex y of G_v has $9 - c(v, y) \in [2, 8]$ neighbours in G_v, $\sum_{y \in G_v} c(v, y) = 72$, and at most 9 vertices y have $c(v, y) = 1$.*

[hand; Lean `regular_card_common`, `regular_nbrIn`, `regular_sum_nbrIn`, `regular_card_private`] MCMC estimates put $|R(3, 9; 30, 99)|$ near 10^{18} (Sec. 6), so case R is not a gluing case.

Lemma P (pair, exact edge count). *For non-adjacent u, w with $C = N(u) \cap N(w)$, $c = |C|$, $A = N(u) - C$, $B = N(w) - C$, $f = e(A, B)$, $X = V - N[u] - N[w]$: $|X| = 38 - d(u) - d(w) + c \le 27$ and $e(X) = e - Z(u) - Z(w) + \sum_{x \in C} d(x) + f \ge e(3, 8, |X|)$. In case R: $|X| = 20 + c$ and $e(X) = 18 + 9c + f$.*

[hand, CONFIRMED Review 13; uses D-E8, F2]

Lemma F. *In case R, for a non-adjacent pair with codegree c: (F2) $9 - c \le f \le (9 - c)^2$ ($C + A' + B'$ is independent for any edgeless $A' \subseteq A$, $B' \subseteq B$, so A–B has a perfect matching); (F3) for every v, $\sum_{y \in G_v} f(v, y) = 648 - \sum_{y \in G_v} c(v, y)^2$, and summed over all non-adjacent pairs, $\sum f = 12960 - \sum c^2$.*

[hand, CONFIRMED Review 24]

Lemma Mc (max-codegree split). *Let c^* be the largest codegree of a non-adjacent pair. Then $3 \le c^* \le 7$, and after relabelling: $N(0) = \{1..9\}$, $0 \not\sim 10$, $N(10) \cap N(0) = \{1..c^*\}$, and $c(u, w) \le c^*$ for all non-adjacent u, w.*

Proof. The 600 non-adjacent pairs have codegree sum $40 \cdot \binom{9}{2} = 1440 > 2 \cdot 600$; $c^* \le 7$ by L6. □

In $R(3)$ all codegrees are 1, 2 or 3 and $n_3 = n_1 + 240$. [hand, CONFIRMED; Lean `Mc.lemma_Mc`] Symmetry breaking must then fix the blocks $\{0\}$, $\{1..c^*\}$, $\{c^*+1..9\}$, $\{10\}$, $\{11..39\}$.

Lemma R7. *If $c^* = 7$ for the pair (u, w), then $f = 4$ (Lemma P gives $f \ge 4$, and $f \le 2 \cdot 2$), $|A| = |B| = 2$ with A–B complete, $e(X) = 85$, and X is one of the 4 graphs of $R(3, 8; 27, 85)$ (D-C8e).*

[hand, CONFIRMED; Lean `R7.r7_count`, `R7.r7_cover`; hypotheses F2 and $e(3, 8, 27) \ge 85$] So "every 9-regular $(3, 10, 40)$-graph has $c^* \le 6$" follows from UNSAT of one cube PRc7 with valid block symmetry breaking (Lean `R7.no_codeg7`, `Rc7.prc7_inv`, `Rc7.lexleader_rc7`). Status of that SAT run: Sec. 7 (464 cubes, 365 UNSAT with verified proofs, 99 open). $R(7)$ is not closed.

Branch $R(6)$, $f^* \le 5$. For $c^* = 6$, choose the codegree-6 pair with the least $f = f^*$. If $f^* \le 5$, X is in $R(3, 8; 26, 75..77)$, a published class of 1,765,856 graphs (D-C8d; $e = 73, 74$ are impossible since $f \ge 3$). Extending each X to G costs ≥ 1 h per X by SAT (all 9 sampled X timed out: $e = 75$: 2 $X > 1$ h, $e = 76$: 1 $X > 2$ h, $e = 77$: 6 $X > 1$ h; logs `runs/cases/r6f/x75s`, `x76s`, `x77s/results.txt`) and ∼40 min per X by two-step gluing, i.e. $\ge 10^2$ CPU-years; we do not pursue this branch (Sec. 8). [measurement]

3.8 Local rules for SAT (all G)

Lemma H. *For an independent s-set I, $X_I = V - I - N(I)$ has $\alpha \le 9 - s$, so $|X_I| \le R(3, 10 - s) - 1$ and $e(X_I) \ge e(3, 10 - s, |X_I|)$ ($s = 1$: L1/L3; $s = 2$: L6).*

[hand, CONFIRMED Review 13; uses D-R1 and the $e(3, k, n)$ tables of [GR]]

Lemma N. *If $d(v) = 9$, every vertex of G_v has a neighbour in $N(v)$, and each x in $N(v)$ has at most one vertex of G_v whose only neighbour in $N(v)$ is x (no MTF needed).*

[hand, CONFIRMED Review 13]

Lemma W. *In cube N we may also assume that vertex 0 has the least $e(G_v)$ among 9-vertices, and $e(G_x) \ge e(3, 9, 39 - d(x))$ for all x; valid only with symmetry breaking that fixes 0 and the blocks.*

[hand, CONFIRMED Review 21]

Excess lemma ($m \ge 13$). *Under sel, MTF, $e \ge 167$ and non-regularity, at least 13 degree-9 vertices have $e(G_v) \le 98$, so two of them are adjacent.*

[cond: sel; hand, CONFIRMED Review 28]

3.9 Further structural approaches

These are reported so that others need not repeat them; details in Sec. 8.

- Counting LP over independent-set profiles for case R (I_s, $|X_S|$ distributions): feasible (CONFIRMED Review 22).
- Excess forcing over the 200 sel distributions: at $T \le 90$ no vertex is forced in any of them (`notes/cases.md` Sec. 7).
- Pair and profile LPs with (F1)–(F3): feasible; (F3) is no contradiction.
- Edge kernels (CONFIRMED Review 30) for R and for the near-regular distributions: the near-regular catalogue alone has $4.4 \cdot 10^7$ unique kernels for codegree ≤ 6 (case R: Sec. 8), and 14 of 16 sampled near-regular kernels exceeded 1 h of SAT (Sec. 8).

4 Formal verification in Lean 4

4.1 Setting

Lean 4 [Lean4] v4.35.0 with Mathlib [Mathlib20] tag v4.35.0 (commit 56a0058); project `lean/` (library `R310`). A build checks every theorem below in the Lean kernel. There is no `sorry`, no new `axiom` and no `native_decide`; `#print axioms` gives only `propext`, `Classical.choice`, `Quot.sound` for all 263 theorems (`scripts/Axioms.lean`). A $(3, 10, 40)$-graph is `G : SimpleGraph (Fin 40)` with `G.CliqueFree 3` and `G.indepNum` ≤ 9; G_v is `G.induce (V \ N[v])`, and MTF is `Maximal (·.CliqueFree 3) G`.

4.2 Hypotheses

Published facts are never proved in Lean. Each theorem that needs one takes it as an explicit argument:

Lean hypothesis	meaning	source
`Ramsey3Le 9 36`	$R(3,9) \leq 36$	D-R3
`Ramsey3Le 8 28` (= `FactF2`)	$R(3,8) \leq 28$	D-R2 (F2)
`FactF1`	$e(3,8,m) \geq 6m - 91$ for $23 \leq m \leq 27$	F1 (D-E8; [cert] for $m = 23, 24, 27$)
`E39Table`	$e(3,9,m) \geq 86, 95, 104, 118, 129, 140$ for $m = 30..35$	D-E9
`EdgeLowerBound 10 40 161`	$e(3,10,40) \geq 161$	D-E10b
`EdgeLowerBound 8 27 85`	$e(3,8,27) \geq 85$	D-E8 (Lemma R7 only)
`Regular35`	every $(3,9;35)$-graph is 8-regular	D-G1 (L7 only)
`Glue U G`	no vertex of degree d has $e(G_v) \leq U_d$, for (d, U_d) in U	gluing runs, Sec. 6 [enum]

4.3 What is formal

1. **All hand lemmas of Sec. 3.1–3.3** (L1–L9, M0–M3, the class-LP rows with weighted Mantel, Theorems A, A″, B, C), with the Lean names given there.
2. **Cover Lemma** (Sec. 3.6) from F1 and F2 only: `cover_lemma`; over SAT assignments `Enc.cover_assign` (with `Enc.fdef_iff`: the clauses TRI + IS(10) hold iff the assignment's graph is a $(3, 10, 40)$-graph); over the cube files `Files.cover_files`. `lake env lean scripts/EmitCubes.lean` writes the cubes as OPB files (99,701 / 99,700 / 99,700 constraints for R / S8 / N), which the SAT pipeline compares with its own formulas.
3. **Case tables** (Sec. 3.4): for pinned outputs of `cases/degseq.py` (`lean/data/cases_pin1`, with SHA-256 sums), the kernel checks by `decide +kernel` that the enumeration and all 19,890 LP certificates are correct: `caseTable_<tag>` (`distOf G` is in the list) for the four tags gr, gr_mtf, self, self_mtf, and `t9_<tag>_<scenario>_main` (given `Glue U G`, some 9-vertex has $e(G_v) \leq t_9$). A Python mirror of the Lean rule set gives the same distribution sets for all four tags.
4. **Case R lemmas**: `Mc.lemma_Mc`, `R7.r7_count`, `R7.r7_cover`, `R7.no_codeg7`, `Rc7.prc7_inv`, `Rc7.lexleader_rc7`.
5. **Symmetry breaking**: the lex-leader lemma `Files.cover_files_sb` (a model of a cube gives a model of the cube plus every valid symmetry-breaking clause whose witness fixes 0 and $\{1..d\}$); the general-n form `Gen.lexleader_gen`, `Gen.lexleader_inv` (any formula invariant under the witnesses) and `Gen.edgeLowerBound_of_unsat` (UNSAT + valid SB clauses give an `EdgeLowerBound`, the form of F1).

6. **Verified checker `sbcheck`**: `SB.checkSB_sound` and `Gen.checkG_sound` prove that a clause accepted by the executable is a valid SB clause in exactly the sense of item 5. Trusted: the input parser (`SBMain.lean`) and the Lean compiler. It prints "`s VERIFIED k clauses`" only if every clause passes (0.66 ms per clause per CPU). It is used for the symmetry-breaking clauses of the certified SAT runs (Sec. 7).

4.4 Scope of the formalisation

The following parts are outside the Lean development.

- The published facts of Sec. 4.2 (hypotheses).
- The gluing enumerations (Sec. 6): they enter only through `Glue U G`.
- The SAT proofs themselves: they are checked by cake_lpr / VeriPB (Sec. 7). Lean covers the link from the mathematical statement to the cube formulas (items 2, 5, 6).
- Lemmas H, P, F, N, W and the excess lemma (Sec. 3.7–3.8): hand proofs, CONFIRMED (Sec. 2.2).
- The 7 Farkas certificates of the sel profile ILP: checked in exact rational arithmetic by a Python script.
- The case tables are checked for the pinned outputs only; a change of `cases/degseq.py` needs a re-pin.

5 Symmetric graphs

Here we ask whether a $(3, 10, 40)$-graph G can have a large or a prime-order symmetry. Every claim has one of these status labels: **proof** (a short hand proof, independently re-checked), **certified** (an independent checker rebuilds the CNF; CaDiCaL 3.0.1 [CaDiCaL24] writes a DRAT proof; drat-trim [DRATtrim14] prints `s VERIFIED`), **enumeration** (a complete search run with no proof log), **open**. Code: `symsat/`. Logs: `runs/symsat/`. Full notes: `notes/symsat.md`. Published facts used: $R(3, k)$ for $k \le 9$ ([DS1, rev. 18, Table I]; [McZ]; [GrR]); the complete list of $(3, 8, 27)$-graphs (477 142 graphs, McKay's data page [McKayWeb]) [McZ]; $\delta(G) \ge 5$ (Sec. 3; Review 4, which relies on [GR] Thm 3) is used only in search runs labelled "dmin 5", never in certificates.

5.1 Elementary facts (proof)

- (F3) G is connected. If $G = G_1 + G_2$ with $\alpha(G_i) = a_i$, then $a_1 + a_2 \le 9$ and $|G_i| \le R(3, a_i + 1) - 1$. So $40 \le \max_a [R(3, a + 1) + R(3, 10 - a) - 2] = 36$; contradiction.
- (F4) No automorphism of prime order $p \ge 11$. Let σ have order p, with fixed set F and non-fixed set M. A vertex in F with a neighbour in M has $\ge p > 9$ neighbours. So no edge joins F and M, and by (F3) F or M is empty. F empty means $p \mid 40$. Thus p is in $\{2, 3, 5, 7\}$, and $f = |F| = 40 \bmod p$.
- (F6, twins) Let O be a σ-orbit of size p, and let $A_O = N(O) \cap F$. Every x in O has $N(x) \cap F = A_O$. If $G[O]$ has an edge, then A_O is empty.
- (F7, fixed-point bounds) Let $f > 0$. Some orbit O has $a = |A_O| \ge 1$, and O is independent. Lemma L ($\alpha(G - N[I]) \le 9 - |I|$ for I independent) with $I = O$ gives $40 - a - p(10 - a) \le R(3, 10 - p) - 1$ and $f - a \le R(3, 10 - p) - 1$. Hence: $p = 7$: $f = 5$; $p = 5$: $f \in \{0, 5, 10, 15\}$; $p = 3$: $f \in \{1, 4, \ldots, 28\}$; $p = 2$: $f \in \{0, 2, \ldots, 34\}$. This leaves **33 cases** (p, f): $1 + 4 + 10 + 18$. (The same numbers appear as "Theorem B" of unsolvedlab.com; our proof is independent.)

5.2 Cayley graphs (certified)

Every group of order 40 is $Z_5 \rtimes P$ with $|P| = 8$ (Sylow + Schur–Zassenhaus). We build all 28 semidirect products $Z_5 \rtimes_\varphi P$ (5 groups P, all $\varphi : P \to Z_4$) and sort them by an explicit isomorphism test into 14 classes $G_0..G_{13}$ (14 is the known number of groups of order 40). The searches run on these 14 representatives. Completeness holds because each removed product is mapped to its representative by an explicit isomorphism, which `symsat/groups.py` checks to be a bijective homomorphism.

Theorem 5.1. *No $(3, 10, 40)$-graph is a Cayley graph.*

- Method A (enumeration): all maximal product-free symmetric sets S, $|S| \leq 9$; exact α. The minimum α is 10 for 12 groups and 11 for 2 groups ($G_1 = Z_5 \rtimes Z_8$ with $|\varphi(P)| = 4$, and $G_{13} = Z_5 \rtimes Q_8$ with $|\varphi(P)| = 2$). < 2 s in total. For Z_{40} (circulants) this agrees with Harborth–Krause [HaKr].
- Method B (certified): orbit SAT (Sec. 5.3) with the left-regular action, no symmetry breaking, no degree clauses. **14/14 DRAT VERIFIED** (`symsat/verify_cert.py`, `runs/symsat/cayley_verify.log`).

5.3 Prime-order automorphisms: method

Fix $\sigma = p$-cycles on $\{0..kp-1\}$ and f fixed points. Then G is σ-invariant. There is one Boolean per σ-orbit of vertex pairs. Clauses: triangles (projected); independent 10-sets added lazily (CEGAR or the satcore IPASIR-UP propagator; each clause is implied by $\alpha \leq 9$); optional degree clauses (implied, for propagation only); lex-leader symmetry breaking for generators of a subgroup of the normalizer of $\langle\sigma\rangle$. This is sound because the constraint set is invariant under the normalizer. Certificates (`symsat/verify_sc.py`): the checker builds its own orbit variables and lex-leader CNF. It checks that each clause of the solver's CNF is a triangle clause, one of its own lex clauses, or a clause that it re-derives from an independent 10-set. Then drat-trim runs. The trust base is drat-trim, `verify_sc.py` (about 200 lines) and the soundness argument for the lex-leader clauses. The solver, its propagator and the degree clauses are not in the trust base. Calibration on complete published families $(3, 6, 17)$, $(3, 7, 22)$, $(3, 8, 27)$, $(3, 9, 35)$: 139 (p, f) cases. 133 cases were decided, with **0 mismatches** against nauty [nauty14] ground truth. Every (p, f) type that occurs in these families was recovered as SAT (for example (35: 2,1) in 1236 s). Every SAT model passes `tools/check_graph.py`. Positive control with the checker's own lex clauses at $n = 35$: $(7, 0)$, $(3, 5)$ and $(5, 0)$ are SAT. Negative control (honest limit): the known $(3, 10, 39)$-graphs with an involution of type $(2, 1)$ or $(2, 3)$ were not found within 4 h (6 runs, `runs/symsat/bench39*.jsonl`). So for $p = 2$ with few fixed points, a TIMEOUT tells us nothing.

5.4 Results for the 33 cases (p, f)

Theorem 5.2 (certified). *A $(3, 10, 40)$-graph has no automorphism of order 7. It has no automorphism of order 5 with $f \in \{10, 15\}$ fixed points. It has no automorphism of order 3 with $f \in \{13, 16, 19, 22, 25, 28\}$.*

(9 cases. Each case was solved UNSAT by satcore in 5–110 s, and each is DRAT VERIFIED.)

p	f	status	evidence
7	5	certified	UNSAT 6.8 s, VERIFIED
5	15, 10	certified	UNSAT 7.2 s, 46 s; VERIFIED
5	5	enumeration	UNSAT 1322 s (no DRAT log)
5	0	open	TIMEOUT 3600 s, and 24 h (dmin 5, `p50_search`)
3	28, 25, 22, 19, 16, 13	certified	UNSAT 5–110 s; VERIFIED
3	10, 7, 4, 1	open	see subcases below
2	34 ... 0 (18 cases)	open	see subcases and Sec. 5.5

Subcase split (proof of coverage; runs = enumeration). Let $f > 0$, and let $a =$ max over orbits of $|A_O|$. Relabel so that orbit 0 is independent with $N(\text{orbit } 0) \cap F = A$, $|A| = a$, and every other orbit has $\leq a$ fixed neighbours. The subcases $a = \max(1, f - R(3, 10-p)+1) .. \min(f, a_{\max}(p))$ cover (p, f). $a_{\max}(p)$ is 7 for $p = 2$ and 6 for $p = 3$. The extra lex-leader generators fix orbit 0, A and $F - A$ setwise; `verify_sc.py` checks this for each generator. Runs: satcore, dmin 5, full transposition lex-leader. First pass 1800 s (115 subcases); rerun of the timeouts 21 600 s (`runs/symsat/sub_search.jsonl`, `rerun1.jsonl`; 73/73 done). A planned 24 h rerun of the 62 subcases that timed out (rerun2) was stopped by a project decision (2026-10-26). Of these 62, 3 ($(2, f, 7)$, $f = 28, 30, 32$) are empty by Lemma 5.3; the other 59 are **open**. Status on 2026-10-26 (U = UNSAT by enumeration, T = TIMEOUT = **open**, F10 = closed by Sec. 5.5, – = not in range):

$(p, f)\backslash a$	1	2	3	4	5	6	7
(3,10)	U	T	T	U	U	U	–
(3,7)	U	T	U	U	U	U	–
(3,4)	T	T	U	U	–	–	–
(3,1)	T	–	–	–	–	–	–
(2,34)	–	–	–	–	–	–	T ($f = 34$: see below)
(2,32)	–	–	–	–	T	T	F10
(2,30)	–	–	U	T	T	T	F10
(2,28)	U	U	U	T	T	T	F10
(2,26)	U	U	U	T	T	T	U, F10
(2,24)	U	U	T	T	T	T	U, F10
(2,22)	U	U	T	T	T	U	U, F10
(2,20)	U	U	T	T	U	U	U, F10
(2,18)	U	U	T	T	U	U	U, F10
(2,16)	U	U	T	T	T	U	U, F10
(2,14)	U	U	T	T	T	U	U, F10
(2,12)	U	T	T	T	T	U	U, F10
(2,10)	U	T	T	T	T	U	U, F10
(2,8)	U	T	T	T	T	T	U, F10
(2,6)	T	T	T	T	T	U	–
(2,4)	T	T	T	T	–	–	–
(2,2)	T	T	–	–	–	–	–
(2,0)	no split ($f = 0$); open						

53 of 115 subcases are UNSAT, 3 more ($(2, f, 7)$, $f = 28, 30, 32$) are empty by Lemma 5.3, and the other 59 are open. Every (p, f) case with an open subcase stays open: $(3, f)$ for $f \in \{1, 4, 7, 10\}$, and all 18 cases of order 2. No (p, f) case is closed by subcases alone. Subcase UNSAT results have no proof logs.

5.5 Order 2, $a = 7$: reduction to McKay's $(3, 8, 27)$ list

Lemma 5.3 (proof; CONFIRMED Review 25). *Let $p = 2$, $O = \{x, x'\}$, $|A_O| = 7$. Then $|N[O]| \le 2+7+2\cdot 2 = 13$. Lemma L gives $|V - N[O]| \le R(3,8)-1 = 27$, so equality holds everywhere. Then $\deg x = \deg x' = 9$, and $H = G - N[O]$ is a $(3, 8, 27)$-graph. $\sigma|_H$ is an automorphism of H of order ≤ 2 with exactly $f - 7$ fixed points.*

In the complete $(3, 8, 27)$ list, the involution types are only $(2, 1)$, $(2, 3)$, $(2, 5)$, $(2, 7)$, $(2, 9)$, $(2, 17)$ (nauty + full enumeration of group elements). Hence:

- the subcase $(2, f, 7)$ is **empty (proof)** for $f \in \{18, 20, 22, 26, 28, 30, 32\}$;
- for $f \in \{8, 10, 12, 14, 16, 24\}$, $(2, f, 7)$ reduces to extending each pair (H, conjugacy class of involutions of type $(2, f - 7)$) by 13 vertices. These are **2557 small SAT problems** ($f = 8$: 2263, 10: 250, 12: 35, 14: 6, 16: 2, 24: 1). **All 2557 are UNSAT (enumeration).** Certificates: **2557/2557 DRAT VERIFIED, 0 failed** (`symsat/ext27_verify.py`: own CNF rebuild, CaDiCaL 3.0.1 + drat-trim; `runs/symsat/ext27_verify.log`, `ext27_verify2.log`). So these 6 subcases are **certified**.
- $f = 34$ ($\sigma|_H = \mathrm{id}$): this needs the extension of all 477 142 graphs. A sample of 11 graphs (every 47 714th) is UNSAT, at about 380 s per graph (`runs/symsat/ext27_f34test.log`), so the full run costs about 50 000 CPU-hours. $(2, 34, 7)$, and so $(2, 34)$, is open.

Trust base: $R(3, 8) = 28$ and the completeness of McKay's $(3, 8, 27)$ list [McZ] (published, trusted by rule); nauty 2.9.3 for $\mathrm{Aut}(H)$.

5.6 Summary

- Certified: no Cayley graph (14 groups); no automorphism of order ≥ 11 (proof) or 7; none of order 5 with $f \in \{10, 15\}$; none of order 3 with $f \ge 13$.
- Certified: the 2557 F10 extensions (subcases $(2, f, 7)$, $f \in \{8, 10, 12, 14, 16, 24\}$).
- Enumeration only: order 5, $f = 5$; 53 of 115 subcases (p, f, a) UNSAT (dmin 5); 3 more are empty by Lemma 5.3 (proof); the other 59 subcases are open.
- Open: $(5, 0)$; $(3, f)$ for $f \in \{1, 4, 7, 10\}$; all 18 cases of order 2. Order 2 with few fixed points is hard for our method: known $(3, 10, 39)$-graphs with an involution are not recovered within 4 h.

6 Gluing exclusions

Status legend: **enumeration-only** = exhaustive computer search without a proof certificate; **CONFIRMED** = independently re-checked with other code (no proof certificate).

6.1 Method

Let G be a $(3, 10; 40)$-graph and v a vertex of degree d. Then $N(v)$ is independent and $G_v = G - N[v]$ is a $(3, 9; 39 - d)$-graph, so $4 \le d \le 9$ ($R(3, 9) = 36$ [D-R3]). Conversely G is obtained from G_v by *neighbourhood gluing* ([GR, Sec. 3], [Ang, Sec. 4]): add v and $u_1..u_d$, join v to all u_i, and give each u_i a neighbour set $\{v\} + S_i$ with S_i independent in G_v; G is triangle-free automatically, and $\alpha(G) \le 9$ iff $\alpha(G_v - \bigcup_{i \in K} S_i) \le 9 - |K|$ for every $K \subseteq \{1..d\}$. So: if the gluer returns no graph for every H in a class $\mathcal{C}$ of $(3, 9; 39 - d)$-graphs, then no $(3, 10; 40)$-graph has a degree-d vertex v with G_v in $\mathcal{C}$.

Code (own, C; `gluing/src/`): `glue.c` (the [GR] algorithm: independent sets sorted by size, pair compatibility through the independent 8-sets, degree and edge pruning, exact α test at the leaf); `glue2.c` ([Ang] Sec. 4.2 variant: maximal independent sets first, then subsets). Isomorphism

rejection: nauty `shortg` [nauty14]. All $n = 40$ runs use only "every vertex degree in $[4, 9]$": **no MTF and no edge assumption**, so the results combine with every case split of Sec. 3.

Validation (all passed; exact counts from [GR] T12–T14, [Ang], McKay [McKayWeb], HoG [HoG23]):

- rebuild from scratch of all $R(3, k; n)$ for $k \leq 6$ (e.g. $R(3, 6; 16) = 2576$, $R(3, 6; 17) = 7$);
- $R(3, 7; 21) = 1{,}118{,}436$ with every e-level = [GR] T12; $R(3, 7; 22) = 191$;
- low-edge classes $R(3, 8; 22, 42) = 21$, $R(3, 8; 23, \leq 51) = 102/8241/356041$, $R(3, 8; 24, \leq 58) = 51/3419/129347$ ([GR] T13);
- `glue.c` and `glue2.c` give identical output sets on all test steps; both rebuild C_{35} and Exoo's $(3, 10; 39)$-graph from their own G_v.

Input classes are generated with the [GR] increment method (G in $R(3, k; n, \leq e_{\max})$ has a vertex v of degree d with $e(G_v) \leq e(3, k-1, n-d-1) + t_d - 1$). The increments t_d are checked by exact enumeration of all integer degree sequences (`tcheck.py`, = [GR] inequality (4)); CONFIRMED (Review 20, G5).

6.2 Results

case	input class (all graphs)	size (= [GR] T14, D-C9a/b)	d	outputs	CPU	status
S8, $e = 95$	$R(3, 9; 31, 95)$ (HoG file)	560	8	0	121 s	enumeration-only; CONFIRMED (independent SAT gluer, 560/560 UNSAT)
S8, $e = 96$	$R(3, 9; 31, 96)$ (own generation)	35,154	8	0	6,962 s	enumeration-only; CONFIRMED (Review 31: all inputs valid and pairwise non-isomorphic; own SAT gluer 100/100 UNSAT per class, sample)
N, $e = 86$	$R(3, 9; 30, 86)$ (HoG file)	1,800	9	0	879 s	enumeration-only; independent sample 200/200 UNSAT
N, $e = 87$	$R(3, 9; 30, 87)$ (own generation)	147,335	9	0	64,481 s	enumeration-only; CONFIRMED (Review 31: all inputs valid and pairwise non-isomorphic; own SAT gluer 100/100 UNSAT per class, sample)

The two generated classes: $R(3, 9; 31, \leq 96) = 35{,}714$ graphs (from $R(3, 8; 24, \leq 58)$, $d = 6$, $t_6 = 3$; 67,012 CPU-s) and $R(3, 9; 30, \leq 87) = 149{,}135$ graphs (from $R(3, 8; 24, \leq 58)$, $R(3, 8; 23, \leq 51)$, $R(3, 8; 22, 42)$, $d = 5, 6, 7$; $\sim$293,000 CPU-s; internal cross-check: the $d = 6$ step alone gives all 149,135, $d = 5$ gives 149,106, $d = 7$ gives 0). Per-e counts equal [GR] T14; the minimum levels ($e = 95, 86$) are the same sets as the HoG files; the files were re-checked independently (pairwise non-isomorphic, triangle-free, $\alpha = 8$; Review 31). Logs: `runs/gluing/final40.log`, `runs/gluing/n40/`. Since $e(3, 9, 31) = 95$ and $e(3, 9, 30) = 86$ [D-E9], these are all graphs below the bounds:

Theorem 6.1 (enumeration-only). *In a $(3, 10; 40)$-graph, every vertex v of degree 8 has $e(G_v) \geq 97$ and every vertex v of degree 9 has $e(G_v) \geq 88$.*

With $e(G_v) = e(G) - Z(v)$ ($Z(v)$ = sum of the degrees of $N(v)$), these are linear constraints on degree distributions; Sec. 3 uses them as "S8 closed" and "N($t_9 \leq 87$) closed" (20 of the 545 MTF distributions are closed by S8 + $t_9 \leq 87$, i.e. $t_9^{U8} \leq 87$; numbers from the case table, Sec. 3).

6.3 How far gluing can go: class sizes (MCMC estimates)

Cost of gluing is set by the number of inputs. We estimate $|R(3,k;n,e)|$ per level with a Markov chain on labelled graphs (edge toggles and edge/non-edge swaps, Wang–Landau weights [WL01], then a transition-matrix estimator $g(e)\langle a\rangle_e = g(e+1)\langle r\rangle_{e+1}$), anchored at an exact HoG level. Checks against 20 published counts: within ∼5% (worst 30%, $n = 33$), e.g. $R(3,9;30,87)$ $1.470\cdot10^5$ (exact 147,335), $R(3,9;31,96)$ 35,160 (35,154), $R(3,9;32,\leq112)$ $1.49\cdot10^9$ (1,554,928,360 [Ang]), $R(3,8;26,\leq77)$ $1.80\cdot10^6$ (1,767,543 [Ang]). Method CONFIRMED (Review 20, G1); a region that the chain does not visit can only make the true counts larger, so the estimates err on the low side.

e	87	88	89	90	91	92	94	96	99
$\lvert R(3,9;30,e)\rvert$ (estimate)	1.47e5	5.9e6	1.5e8	2.8e9	4.1e10	4.9e11	5.0e13	3.5e15	9.5e17

Consequences (cost per input measured on this machine: generation ∼0.5–1 s, 9-regular gluing 3 ms; [Ang]'s code is ∼4 ms for both):

- $t_9 \leq 88$: $\sim6\cdot10^6$ graphs, ∼4 days on 32 cores. $t_9 \leq 90$ needs ∼0.9 CPU-year even at [Ang] speed; $t_9 \leq 92$ ∼80 CPU-years.
- The 9-regular case: every G_v is in $R(3,9;30,99)$, $\sim9.5\cdot10^{17}$ graphs, i.e. $\sim9\cdot10^7$ CPU-years for the gluing alone. Gluing cannot close it.
- Two-step gluing for sub-case $R(6,f^*\leq5)$ (X in $R(3,8;26,75..77)$, $1.77\cdot10^6$ graphs): measured ∼40 min per X (462k intermediate graphs per X), ∼130 CPU-years. Not feasible.

Our code is ∼100× slower than [Ang] for generation; tuning (α table, clique-colouring bound, edge-budget index ranges; `glue5`, same output as glue) gave 1.2×.

7 Certified SAT computations

Status labels as in Sec. 5: **certified** = every step is re-checked by a checker that does not trust the search code (see 7.2); **enumeration** = a complete solver run with no checked proof; **open**. Code: `sms/` (solver build, encoder, scripts), `cert/` (checkers, glue). Full notes: `notes/sms.md`, `notes/cert_design.md` (v3; Reviews 11, 14, 17, 18), `notes/core_plan.md`.

7.1 Solver: SAT Modulo Symmetries with a lazy independent-set propagator

We use SAT Modulo Symmetries (SMS) [KS24]: CaDiCaL (IPASIR-UP fork 2.1.3) plus a minimality propagator. During search it tests whether the partial graph can still be lex-minimal (row-wise upper-triangle edge vector, $0 < 1$). If not, it adds a clause together with a permutation witness. Build: `github.com/markirch/sat-modulo-symmetries` commit 63958bd, with our patches (`sms/sms-src`):

1. `-forbid-independent-set k`: a lazy propagator for $\alpha < k$. On each call it searches a k-clique in the graph of pairs assigned "non-edge" (bitset branch and bound) and adds the clause "some pair of this k-set is an edge" (45 positive literals for $k = 10$). It also runs on every full assignment, so it is exact. For $(3,10,40)$ the eager encoding has $\binom{40}{10} = 847{,}660{,}528$ clauses; the lazy one adds only the clauses the search needs.
2. `-proof-log`: a CaDiCaL tracer that logs every original clause (input clauses AND clauses added by the propagators) with its ID, and every derived clause with LRAT hints. The stock `-lrat-output` omits the external clauses, so its LRAT does not check.
3. Cube support: `-cutoff-edge-vars-only`, `-lookahead-only-edge-vars`, `-cube-only-decisions`.

Encoder `sms/gen_ramsey.py`: triangle clauses, Sinz counters for degree/edge bounds, MTF, $N(0) = \{1..d\}$ (with `-initial-partition`, so SMS uses only partition-preserving permutations).

Validation (enumeration; 1 CPU; every output graph checked by nauty `shortg` and `tools/check_graph.py`): SMS reproduces McKay's counts of all $R(3,k;n)$ graphs for $k \leq 6$ (e.g. 275,086 for $(3,6,13)$), $R(3,7;21) = 1{,}118{,}436$ (743 s), $R(3,7;22) = 191$, and 0 for $(3,5,14)$, $(3,6,18)$, $(3,7,23)$. Calibration of the cube-R shape: $(3,8,28)$ 7-regular, MTF, $N(0)$ fixed: UNSAT 1846 s; $(3,9,36)$ 8-regular, MTF, $N(0)$ fixed: UNSAT 724 s. (8-regularity of a $(3,9,36)$-graph is a known lemma; this is a calibration, not a new proof of $R(3,9) = 36$.) Limit: SMS canonicity makes it slow at FINDING graphs ($(3,8,27)$: first graph 431 s with SMS, 1.3 s without), so we use it only for UNSAT.

7.2 Certificate chain

No step trusts the search code (CaDiCaL, SMS, the cube generator). The search gives hints; checkers re-check them. For one problem $P = (\Phi + \text{cube constraints} + \text{symmetry breaking } SB)$, split into cubes c_j:

link	statement	checked by
(a) encoding	the input CNF says what the cube says	Lean cube file (`lean/out/cube_*.opb`, from the Lean Cover Lemma, Sec. 4) → SMS base CNF by a VeriPB [VeriPB23] proof (aux definitions by `red`, Sinz counters by `pol`/`rup`), checked by VeriPB and by CakePB [CakePB24]; `cert/base40_cert.py` also regenerates the SMS base exactly
(b) lazy IS clauses	each clause added by the IS propagator is a valid 10-set clause	`cert/check_cnf.py` (exact shape: 10 vertices, all 45 pairs, positive); it also checks that every triangle clause is present
(c) SB clauses	each SMS minimality clause is valid under the one lex order	Lean-verified checker `sbcheck` (`R310.SB.checkSB_sound`, with block partitions `Gen.checkGB_sound`) + Lean lex-leader lemma (`R310/LexLeader.lean`, `Gen.lexleader_inv`); `cert/check_sb.py` as a second, independent check
(d) cover	$\Phi_{\text{base}} + \neg c_1 + \cdots + \neg c_m$ is UNSAT	one SMS run with proof log → LRAT → cake_lpr [cakelpr21]
(e) cubes	$\Phi_{\text{base}} + \text{units}(c_j) + L_j + SB_j$ is UNSAT for each j	per cube: SMS proof log → LRAT → cake_lpr; (b) and (c) on that cube's L_j and SB_j

Verdicts are read from stdout only ("`s VERIFIED ...`"): cake_lpr and CakePB exit 0 also on failure, and cake_lpr's range mode prints "`VERIFIED INTERVALS COVER`" also without an empty clause; the glue tests for that. VeriPB/CakePB proofs of SB were run on samples only; the full SB lists are checked by `sbcheck` (route (c)), which needs no VeriPB. Trusted: the Lean kernel and compiler, `sbcheck`'s 100-line parser, cake_lpr and CakePB (verified checkers), `check_cnf.py` (about 120 lines), the glue scripts, and the hypotheses of the Lean lemmas (published facts, Sec. 2). Versions: cake_lpr a36874a, CakePB 438b8f8, VeriPB 3.0.2, drat-trim 2e3b2dc, CaDiCaL 3.0.1, Lean 4 v4.35.0 (full list with commits in `tools/README.md`). Negative tests (cert): a reversed SB clause, a removed IS clause, a changed triangle clause, a shortened SB clause and a summary without empty clause are all rejected. Positive controls at $n-1$ ($(3,5,13)$, $(3,6,17)$) find the known graphs with the same SB and cubes.

7.3 Certified edge bounds

Each row: "no $(3,k,m)$-graph with at most E edges", degree $\leq k-1$ implied, SMS UNSAT with proof log; then the full chain of 7.2 (one cube). F1 is the hypothesis $e(3,8,m) \geq 6m - 91$,

$23 \le m \le 27$, of the Lean Cover Lemma (Sec. 3–4). The published values are larger ([GR] Table 3 / 13, D-E7/D-E8); these runs make three of the five F1 instances independent of the literature.

statement	SMS UNSAT	lazy IS / SB clauses checked	cake_lpr	status
$e(3,7,17) \ge 25$ (no graph with ≤ 24 edges)	1.6 s	1,292 IS; 851 SB (sbcheck 0.1 s)	14.3 s, VERIFIED	certified (re-run 2026-10-26)
$e(3,7,18) \ge 30$ (≤ 29)	4.4 s	1,975 IS; 1,588 SB (sbcheck 0.2 s)	18.2 s, VERIFIED	certified (re-run 2026-10-26)
$e(3,7,19) \ge 37$ (≤ 36)	24.3 s	3,871 IS; all SB by full VeriPB + CakePB (no sbcheck)	VERIFIED	certified
$e(3,7,20) \ge 44$ (≤ 43)	151 s	7,410 IS; all SB by full VeriPB + CakePB (no sbcheck)	VERIFIED	certified
F1, $m = 23$: $e(3,8,23) \ge 47$ (≤ 46)	3,094 s	27,522 IS; 71,031 SB (sbcheck 10.4 s)	403 s, VERIFIED	certified
F1, $m = 24$: $e(3,8,24) \ge 53$ (≤ 52)	8,970 s	45,786 IS; 112,512 SB (18.4 s)	898 s, VERIFIED	certified
F1, $m = 27$: $e(3,8,27) \ge 71$ (≤ 70)	5,503 s	52,044 IS; 65,321 SB (12.6 s)	701 s, VERIFIED	certified
F1, $m = 25, 26$	–			published [GR]

Logs: `runs/cert/f1/cert_*.log` ("`RESULT: OK ... certified`"), `runs/sms/f1_*.log`, `runs/sms/emin_*.log`. These bounds are weaker than the published values ($e(3,8,23..27) = 49, 56, 65, 73, 85$); they are exactly what the Cover Lemma needs. An edge bound is not implied by Φ, so these runs use pure vertex-permutation witnesses and no MTF (`check_pbp.py -perm-only`; `notes/cert_design.md` Sec. 1.2).

7.4 The sub-case $R(7)$: partial certified result

Setting (Sec. 3: Lemma Mc, Lemma R7; Lean `R7.r7_count`, `R7.r7_cover`, `Rc7.prc7_inv`; hypotheses F2 and $e(3,8,27) \ge 85$ [D-E8]). In the 9-regular cube R, let a non-adjacent pair $(0, 10)$ have the maximum codegree $c^* = 7$. Then the formula PRc7 is: Φ + cube R (9-regular, MTF, $N(0) = \{1..9\}$) + $N(10) \cap \{1..9\} = \{1..7\}$ + the 6 edge units 10-11, 10-12, 8-11, 8-12, 9-11, 9-12 (A–B complete) + all non-adjacent codegrees ≤ 7 + $e(\{13..39\}) = 85$. SMS runs with the block partition $\{0\}$, $\{1..7\}$, $\{8, 9\}$, $\{10\}$, $\{11, 12\}$, $\{13..39\}$. If PRc7 + SB is UNSAT, then every 9-regular $(3, 10, 40)$-graph has all non-adjacent codegrees ≤ 6 (`R7.no_codeg7`).

- Base CNF `sms/cnf/Rc7b.cnf` (444,210 variables, 893,163 clauses): derived from Lean's cube R + the case constraints by VeriPB, checked by VeriPB and CakePB (2.4 min, `runs/cert/base_Rc7b.log`: RESULT OK).
- Cubes: 464 cubes from SMS lookahead (`sms/cubes/Rc7b_L300.cubes`).
- Cover (link d): UNSAT in 14 s; 4,214 lazy IS clauses + 684 SB clauses; cake_lpr "`s VERIFIED UNSAT`"; sbcheck "`s VERIFIED 684 clauses`" (coarse blocks and fine blocks); check_cnf OK. **certified** (`runs/sms/Rc7b_cover/`; full re-check 2026-10-26: "`s VERIFIED cube`").
- Cubes (link e), `runs/sms/Rc7b_cert/SUMMARY.txt`: 6 h limit per cube, 38 CPUs, 18.9 h wall. **365 of 464 cubes UNSAT and certified** (each: cake_lpr + sbcheck with both block partitions + partition test + check_cnf). 99 cubes reached the 6 h limit (**open**). No cube was SAT. CPU per solved cube: median 5.2 s, mean 506 s, p90 256 s, p99 16,910 s, max 20,372 s. Total solve CPU 26 d 17 h (a lower bound: timeouts counted at 6 h); checks 5 h 43 min.
- Re-split of the 99 timeouts (lookahead at 360 assigned edge variables): 37 parents (1 kept whole) → 12,409 sub-cubes; 62 parents (6 kept whole) → 1,668,706 (`runs/sms/rc7_splitA.log`,

`rc7_splitB.log`). In 9.3 h on 31 CPUs, 37 sub-cubes were UNSAT and certified; 28 of the 31 sub-cubes started ran longer than 8 h. So one more split level does not make these cubes easy. We stopped the run at this point.

Result (certified, partial). For 365 of the 464 cubes c_j of a certified cover of PRc7, PRc7 + SB + c_j is UNSAT. $R(7)$ as a whole is **open**: 99 cubes remain. So the bound "codegree ≤ 6" in case R is open.

8 What remains and why it is hard

Here we list what stays open after Sec. 3–7, and what each known route costs. The numbers are our own estimates; we give the method for each one. Most of them are negative results. Status labels as in Sec. 5–7; **estimate** = an extrapolation from samples, not a bound.

8.1 What is open

By the Cover Lemma (Sec. 3–4), a $(3, 10, 40)$-graph gives a model of one of these cubes (Sec. 3, `notes/core_plan.md` Sec. 0):

- **R**: 9-regular, $N(0) = \{1..9\}$ (MTF is implied). Sub-cases $R(c^*)$, $c^* = 3..7$, by the maximum non-adjacent codegree (Lemma Mc). $R(7)$ is partly done (Sec. 7.4: 365/464 cubes certified, 99 open).
- **N**(e), $e = 162..179$: MTF, degrees 6..9, $d(0) = 9$, $N(0) = \{1..9\}$, $e(G) = e$, $88 \leq e(H_0) \leq t_9^{\max}(e)$. After Theorem 6.1 ($T^* = 87$): $e = 161$ is closed; 18 cubes with 524 degree distributions stay (filter of `cases/core_cubes.json`, scenario U8+N86, kept if $t_9 \geq 88$; Review 37). The next gluing level, $t_9 \leq 88$ (∼4 days on 32 cores, Sec. 6.3), closes further distributions if it has no outputs. With the full gluing plan ($T^* = 90$) only $e = 167..179$ (13 cubes, 208 distributions) would stay, but $T^* = 90$ costs about 0.9 CPU-year (Sec. 6.3).
- S8 is closed (Theorem 6.1).

8.2 SAT cube-and-conquer (SMS): cost estimates

Method: Knuth random probes [Knuth75] (`sms/knuth_probe.py`). At level l: cube the formula by lookahead at $C_l = C_0 + 60l$ assigned edge variables, pick one cube at random, solve it (300 s limit); on a timeout, go one level deeper. The estimate $c_0 + b_0(c_1 + b_1(\cdots + b_L t_L))$ is unbiased but heavy-tailed. 1 CPU per probe. Data: `runs/sms/probes/*.jsonl`.

formula	probes	median (CPU-s)	mean (CPU-s)	range of single estimates
cube R, $C_0 = 280$	67	8.0e4	8.4e14 (2.7e7 CPU-years)	2e2 .. 4e16
calibration C1 ($e(H_0) \leq 86$)	30	1.3e7	5.0e10	4e2 .. 1e12
$R(7)$	6	6.3e5	1.9e7	2e2 .. 1e8
$R(6)$	3	6.0e7	1.9e12	1e7 .. 6e12
$R(4)$	1	7.5e8		
$R(3)$	21	7.6e5	7.7e9 (14% of probes > 1e9)	1e2 .. 2e11
$R(5)$	0 finished			

Reading. The tail decides the total. 15 of the 67 R probes (22%) give more than 10^9 CPU-s, so the total for R is very likely at least 10^9 CPU-s (30 CPU-years). Calibration: the certified $R(3, 9)$ computation of Li, Duggan, Bright, Ganesh (IJCAI 2025, arXiv:2502.06055 [DuLBG]) took about 3,300 CPU-h $= 1.2 \cdot 10^7$ CPU-s. The single R estimates go from $2 \cdot 10^{-5}$ to $4 \cdot 10^9$ times that value;

the median ($8 \cdot 10^4$) is below it, and the mean is about $7 \cdot 10^7$ times it. **estimate**: plain SMS cube-and-conquer on cube R is not feasible with our resources.

$R(7)$ shows the spread of a 6-probe estimate. The probe mean was $1.9 \cdot 10^7$ CPU-s. The real certified run used $2.3 \cdot 10^6$ CPU-s ($1.8 \cdot 10^5$ on the 365 solved cubes, the rest on the 99 cubes stopped at 6 h), and the full total is larger (Sec. 7.4). Re-splitting the open cubes made them no easier: 28 of 31 sub-cubes started ran more than 8 h. So the hard cubes are hard as a whole, not only "too large".

Extra constraints tested (all estimate, Knuth probes, same method):

- $R(6)$ with "every codegree-6 pair has $f \geq 6$" (Lemma F, equivalently $e(X) \geq 78$): 3 finished probes 8.0e9, 1.4e11, 1.7e13 CPU-s against 5.6e12, 1.4e7, 6.0e7 for $R(6)$ without it: no gain seen. (A fourth batch was stopped at ∼480 GB RAM; since then smsg runs under `ulimit -v`.)
- Split by maximum degree for F1 $(8, 23, 46)$: $D = 5$ alone took 3,991 s against 3,094 s for the whole formula.
- Cube hardness predictor (Heule's Schur-5 method [Heule18]): on the solved $(3, 9, 36)$ cube sets, static features (free edges, binary triangle clauses, true edges after the cube) have Spearman ρ between -0.51 and $+0.19$ with the solve time. The top 5% by any feature hold at most 3% of the time and miss the tail cube (1060 s of 1855 s). Structural cubes on the $N(0)$–H_0 incidence (empty-hexagon style) were $\geq 10\times$ slower than plain lookahead (1577 cubes, $\geq$ 4489 s with 1 timeout, against 446 s). Neither helps.
- satcore (CaDiCaL 3.0.1 + lazy IS + static lex-leader clauses) and plain CaDiCaL: no answer within 1 h on $(3, 8, 28)$ $D = 7$ and on 8-regular $(3, 9, 36)$, where SMS needs 1846 s and 724 s. SMS's dynamic symmetry breaking is the main gain; we know of no stronger general engine.

8.3 Gluing

Gluing (Sec. 6) is exact but its cost is the number of input graphs. MCMC class sizes (Sec. 6.3): the 9-regular case needs all of $R(3, 9; 30, 99)$, about $9.5 \cdot 10^{17}$ graphs (about $9 \cdot 10^7$ CPU-years). Pair gluing (glue two non-adjacent vertices u, w; $X = G - N[u] - N[w]$ in $R(3, 8; 20 + c)$): worst case, every pair class has at least $\sim 10^{14}$ graphs; on 954 MCMC samples of $R(3, 9; 30, 99)$ the cheapest pair class has median $10^{11.5}$ graphs and only 3.4% of samples have a pair class $\leq 10^{10}$ (`notes/gluing.md` Sec. 8). Two-step gluing of $R(6, f^* \leq 5)$: about 130 CPU-years. **estimate**: gluing cannot close R; it is the right tool only for N(e) with small t_9.

8.4 Kernels and other ideas

Edge-kernel reduction (CONFIRMED, Review 30). Let G be in case R, let M be the maximum codegree of a non-adjacent pair, and for an edge uv let $A = N(u) - v$, $B = N(v) - u$ and $F = G[A, B]$. Then $d_F(a) = c(a, v) - 1 \leq M - 1$. König's theorem gives $\alpha(F) = 16 - \nu(F) \leq 9$, so $\nu(F) \geq 7$. Codegree-1 pairs inside each $N(w)$ form a matching, so there are at most 160 of them. A convexity bound on $\sum c(c-1) = 4C_4$, together with $\sum$ over edges of $e(F_{uv}) = 4C_4$, shows that some edge has $e(F) \leq \lfloor (64 + 20M)/9 \rfloor$, which is 13, 16, 18, 20, 22 for $M = 3..7$. So every graph in $R(M)$ completes one kernel from a finite list. We fix the 18 kernel vertices one by one; SMS permutes the other 22 (cells 1^{18}, 22) and must enumerate all completions (`smsg -a`, `-cutoff 0`, 9-regular, MTF, codegree $\leq M$, lazy $\alpha \leq 9$). Optional: static lex-leader clauses $x \leq_{\text{lex}} x \circ \sigma$ for up to 300 automorphisms σ of the kernel, in the SMS order (lex-min of the row-wise edge vector). Code: `duo/kernel_run2.py`, `duo/kfilt.c`, `duo/ksample.py`.

Catalogues (**enumeration**, genbg 8+8 with $-D(M-1)$, filter $\nu \geq 7$; seconds to minutes of CPU): $M = 3$: 238; $M = 4$: 96,135; $M = 5$: 2,852,820; $M = 6$: 26,326,286 kernels.

Control (**enumeration**): the 6-regular $(3, 7, 22)$ analogue ($M = 5$, 474 kernels) gives exactly the 10 known 6-regular $(3, 7, 22)$-graphs, both with and without the Aut clauses (116 s and 113 s in total; labelg compare).

Sample runs, 2026-10-26 (**estimate**): 32 kernels per M, uniform reproducible sample, 1 h limit per kernel, 1 CPU each, without / with Aut clauses. No completion was found.

M	kernels	solved in 1 h (no Aut / Aut)	solved times (s)	CPU-s lower bound for $R(M)$
3	238	11 / – of the first 15 (old runner; UNSAT is valid)	0.6–33	238 kernels; 4 of 15 > 1 h
4	96,135	2 / 3 of 32	5, 16 / 5, 19, 1782	$\geq 3.2 \cdot 10^8$
5	2,852,820	0 / 0 of 32	–	$\geq 1.0 \cdot 10^{10}$
6	26,326,286	1 / 1 of 32	1.7 / 1.1	$\geq 9.2 \cdot 10^{10}$

The lower bound is N_M times the sample mean, with every timeout counted as 3600 s. Proof logging adds a factor 1.7–2.6 (Sec. 7). The Aut clauses gave no measurable gain. The sum over $M = 5, 6$ alone is $\geq 1.0 \cdot 10^{11}$ CPU-s ($\geq$ 3,200 CPU-years): about 18 times the full budget of 180 CPU-years ($5.7 \cdot 10^9$ CPU-s).

35-vertex control (**enumeration**, partial run): 8-regular $(3, 9, 35)$-graphs, $M \leq 4$ (3,305 kernels), whose answer is known (only $C_{35}(1, 7, 11, 16)$, [GR] Thm 3). Of the first 26 kernels, 11 were UNSAT (0.8–3,550 s) and 15 timed out at 1 h. We stopped the run after 16 CPU-h. The codegree-5 branch (X fixed to one of the 7 $(3, 7, 22)$-graphs with 60–61 edges) was fast: 7/7 UNSAT in 10–18 s. So even one level down, the method does not classify a known case within a day.

Verdict (**estimate**): the reduction is exact. But the hard tail of the completions is the same as in plain SMS cube-and-conquer (Sec. 8.2): each kernel is a hard search. The kernel route cannot close R with our resources.

Other ideas tested (`notes/duo.md`), none of which gives a cut:

- J9 seed (every H_v contains an induced $7K_1 + K_2$; [GR] $e(3, J_9, 30) = 117$): true, but it only breaks symmetry; seeded gluing classes are still 1–4% of $R(3, 9; 30, 99)$.
- Matching lemma (A–B has a perfect matching for non-adjacent u, v in R): true; no SAT gain, because a Hall violation is an independent 10-set, which the lazy propagator already finds.
- Fractional regular-gluing LP on H in $R(3, 9; 30, 99)$ (weights on independent 8-sets): on 1,209 adversarial H (C_{35} minus an independent 5-set minus an edge), the 7- and 8-set rows leave 110 feasible. With degree equalities and all independent 4..8-sets it is infeasible for all 1,209 (LP solver only, no exact certificate). Minutes per H, so it can help only as a uniform certificate.
- Uniform type-based dual of that LP: with coarse types it is provably too weak (exact rational counterexample $H = C_{35} - \{0, 2, 4, 6, 12\}$ − edge $\{19, 20\}$). Fine (WL) types cannot be checked over a finite list. A 6 h solver run on 11 H gave no answer. Route closed.
- Angeltveit–McKay excess / pair overlay: each non-regular "sel" distribution has ≥ 13 degree-9 vertices with $e(H_v) \leq 98$ (CONFIRMED, Review 28). But that catalogue has about $2 \cdot 10^{17}$ graphs, and the 9-regular case has no fixed-sum excess. No cut.
- Codegree-typed pair LP: feasible, $C_4 \in [540, 1050]$, every codegree class may be empty. No cut.
- Split by $q(v)$ = number of C_4 through v . $q(v) = 54 + h_1 + h_4 + 3h_5 + 6h_6 + 10h_7$ (h_c = number of non-neighbours with codegree c), $h_7 \leq 1$, $h_6 \leq 2$ (from $R(3, 7) = 23$ and $R(3, 6) = 18$), $54 \leq \max q \leq 87$, $C_4 \leq 860$. The bounds use $h_1 \leq 9$, $\sum_v h_1 \leq 320$, and "$h_7 > 0 \Rightarrow h_1 \leq 7$" (from Lemma R7, so they use F2 and D-E8); without the last rule, $\max q \leq 88$ and $C_4 \leq 870$. Status: hand proof, CONFIRMED (Review 37: 769 candidates). The $Q = 54$ cube still has 49 kernels.

If $h_6(v) = 2$, then $G - N[\{v, a, b\}]$ is one of the 191 $(3, 7, 22)$-graphs. A probe of this seed gave 6 UNSAT (0.7–277 s) and 3 timeouts (30 min) out of 9: the same heavy tail.

8.5 Summary

route	target	cost	status
SMS cube-and-conquer	cube R	$\geq 10^9$ CPU-s likely; probe mean $8.4 \cdot 10^{14}$	estimate: not feasible
SMS cube-and-conquer	$R(7)$	$>$ 26.7 CPU-days; 99 cubes open after 6 h each	365/464 certified; open
SMS + Lemma F rows	$R(6)$	no gain in 3 probes	estimate
gluing	R (9-regular)	$\sim 9.5 \cdot 10^{17}$ inputs, $\sim 9 \cdot 10^7$ CPU-years	estimate: not feasible
gluing	$t_9 \leq 88$ / ≤ 90 / ≤ 92	~4 days / ~0.9 / ~80 CPU-years	estimate
pair gluing	R	$\geq \sim 10^{14}$ X per pair (worst case)	estimate: not feasible
edge kernels + SMS	R	$\geq 10^{11}$ CPU-s for $M = 5, 6$ (most sampled kernels $>$ 1 h)	estimate: not feasible

What would change this: a new structural lemma that cuts cube R by many orders of magnitude (for example codegree ≤ 6 in R, which needs the 99 open $R(7)$ cubes, or a strong bound on $e(H_v)$ at every vertex), or a much faster engine.

9 Data and code availability

Code, notes and small data files are at https://github.com/jeffchen006/Ramsey-R3-10 (this report: DOI 10.5281/zenodo.23678020). The repository contains:

- Code: our patches to SMS (third-party code by Kirchweger and Szeider; patches released under its licence terms: lazy independent-set propagator, cube and proof support), the gluers (glue, glue2, glue5), symsat's orbit-SAT code, the satcore encodings, the certificate checking scripts, the kernel and case-split encoders, and the Lean 4 development.
- Notes: the technical notes cited in this report (`notes/`, including `notes/critic.md` and `notes/DEPENDENCIES.md`).
- A manifest (`MANIFEST.tsv`) with the SHA-256 hash of every file, and a list of the files left out because of size (`LARGE_FILES.tsv`).

The run logs cited as `runs/`, the large graph classes ($R(3, 9; 31, \leq 96)$, $R(3, 9; 30, \leq 87)$), the cube sets and the large DRAT/LRAT/VeriPB proofs are not in the public repository because of their size; they are available from the author on request. Third-party data (House of Graphs, McKay's lists, published counts) and third-party software are cited, not redistributed; versions and sources are listed in `tools/README.md`.

10 Use of AI, and work in progress

How this report was produced. Most of this work was produced by AI. A team of AI agents (Claude Code, by Anthropic) wrote the code, ran the computations, wrote the technical notes and wrote most of this text, under the direction of the author. OpenAI Codex was used as a research assistant for ideas; the edge-kernel reduction (Sec. 8.4), the split by $q(v)$ and the other ideas listed in Sec. 8.4 came from it (`notes/duo.md`). The author set the goals and chose the directions.

What "CONFIRMED" means. The label CONFIRMED and the review numbers ("Review n") refer to a separate AI agent that re-derived each claim, where possible with its own code; its log is `notes/critic.md`. No human referee has checked the hand proofs or the enumerations. The **certified** results do not depend on this review: they rest on verified checkers (cake_lpr, CakePB, the Lean kernel) and on the published facts of Sec. 2.3. The author does not guarantee results that are not marked as certified.

Work in progress at the time of writing.

- Gluing $t_9 \leq 88$ (`gluing/src/t9_88.sh`, started 2026-10-25): it regenerates $R(3,7;15..17)$ from all of $R(3,6)$ (= [GR] T12), $R(3,8;23,\leq 52) = 10{,}691{,}100$ and $R(3,8;22,\leq 44) = 73{,}895$ (every e-level = [GR] T13), then $R(3,9;30,\leq 88)$ with increments $t_5 = 3$, $t_6 = 4$, $t_7 = 3$, then the degree-9 gluing of $R(3,9;30,88)$. Built-in check: the $e \leq 87$ part must equal [GR] T14. Expected end: 2026-10-29. If it has no outputs, every degree-9 vertex has $e(G_v) \geq 89$.
- Symmetry case $(p,f) = (5,5)$: the solver run is UNSAT, but its DRAT log was lost in a crash; the certified re-run is pending.
- The 99 open cubes of $R(7)$ (Sec. 7.4) and the case $(2,34)$ (Sec. 5.5) were not continued because of their cost.
- F1 for $m = 25, 26$ was not run to the end; these two values rest on the published tables (D-E8).

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